



**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**  
**General Certificate of Education Advanced Level**

**PURE MATHEMATICS**  
**PAPER 2**

**6042/2**  
**3 hours**

**NOVEMBER 2024 SESSION**

Additional materials:  
Answer booklet  
List of Formulae MF 7  
Scientific calculator [Non-programmable]

**INSTRUCTIONS TO CANDIDATES**

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Answer **all** questions in Section A and any **five** questions from Section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets [ ] at the end of each question or part question.

**This question paper consists of 5 printed pages.**

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## Section A (40 marks).

Answer **all** questions in this section.

1 Express  $\frac{2x^2}{2x^2-x-1}$  in partial fractions. [6]

2 Evaluate  $\int_0^1 e^{2x} \cos x dx$ , giving the answer correct to 3 significant figures. [6]

3 (a) Expand  $(3-x)^{-\frac{1}{2}}$ , up to and including the term in  $x^3$  simplifying the coefficients. [6]

(b) State the range of values of  $x$  for which the expansion is valid. [2]

4 (a) Solve the equation

$$x^3 - 4x^2 + x = -6 \quad [5]$$

(b) Sketch the graph of

$$y = x^3 - 4x^2 + x + 6 \quad [2]$$

(c) Hence or otherwise solve the inequality

$$x^3 - 4x^2 + x + 6 < 0 \quad [1]$$

5 A function  $h$  is defined by  $h: x \rightarrow \frac{2}{x-3}, x > 3$ .

(a) Find  $h^{-1}(x)$  stating its domain. [4]

(b) Hence find the value of  $hh^{-1}(2)$ . [3]

(c) Sketch the graphs of  $y = h(x)$  and  $y = h^{-1}(x)$  on the same axes. [3]

(d) State the relationship between  $h(x)$  and  $h^{-1}(x)$ . [2]



## Section B (80 marks)

Answer any five questions from this section. Each question carries 16 marks.

- 6 (a) (i) Express  $2\cos\theta - \sin\theta$  in the form  $R\cos(\theta + x)$  where  $r > 0$  and  $0^\circ < x < 90^\circ$ . [4]
- (ii) Hence or otherwise solve the equation  $10\cos\theta - 5\sin\theta = 3$  for  $0 \leq \theta \leq 360^\circ$  [5]
- (b) Find the maximum and minimum values of  $\frac{1}{2\cos\theta - \sin\theta + 11}$ . [2]
- (c) Given that  $\sin A = \frac{3}{5}$  for  $0 < A < 90^\circ$  and  $\sin B = \frac{\sqrt{3}}{3}$  for  $180^\circ < B < 270^\circ$ , find the exact value of  $\tan(A - B)$ . [5]
- 7 (a) Express  $Z = 1 + i$  in trigonometrical form. [3]
- (b) Find  $(1 + i)^{\frac{1}{5}}$  in trigonometrical form. [8]
- (c) Show on an Argand diagram the region that satisfies  $0 < \arg(Z + 1 + i) < \frac{\pi}{3}$ . [5]
- 8 The eighth term of an arithmetic progression is twice the third term. The sum of the first eight terms is 39.
- (a) Find the
- (i) first term and the common difference. [5]
- (ii) ninth term of the progression in its simplest form. [2]
- (b) Show that the sum of first  $n$  terms of the progression is  $\frac{3}{8}n(n + 5)$ . [2]
- (c) Hence prove by induction that
- $$\sum_{r=1}^n U_r = \frac{3}{8}n(n + 5).$$
- [7]

- 9 The position vectors  $P, Q, R$  of three points are
- $$\begin{aligned} p &= i + j + k \\ q &= i + 2j + 3k \\ r &= i - 3j + 2k \end{aligned}$$

respectively

- (a) Find a unit vector parallel to  $p + q + r$ . [6]
- (b) Show that the cosine of the angle between  $p + q + r$  and the vector  $p$  is  $\sqrt{\frac{3}{5}}$ . [5]
- (c) Hence find the vector equation of a plane through  $P, Q, R$ . [5]

- 10 (a) Prove by mathematical induction that

$$\sum_{r=1}^n \frac{2r+1}{r^2(r+1)^2} = 1 - \frac{1}{(n+1)^2}$$

for all positive integral values of  $n$ . [7]

- (b) The set  $S = \{a; 2; 4; 6\}$  forms a group under the operation addition modulo 8.

(i) Write down the value of  $a$ . [1]

(ii) Construct a Cayley table for  $S; \tau_8$ . [2]

- (c) Show that the set of functions of the form  $f(x) = px + q$ , where  $p, q \in \mathbb{R}$  and  $p \neq 0$ , forms a group under composition of functions. [6]

- 11 (a) (i) Express  $y = x^2 - 2x - 3$  in the form  $(x - p)^2 - q$ , where  $p$  and  $q$  are constants. [2]

(ii) State the values of  $p$  and  $q$ . [2]

- (b) (i) Hence or otherwise sketch the graph of  $y = x^2 - 2x - 3$ , showing clearly the intercepts and turning points. [3]

(ii) State the range of values of  $x$  for which  $y$  is negative. [1]

- (c) (i) If  $y = 3^x$ , show that the equation  $3^{2x} + \frac{1}{3^{-x}} = 2$ , can be written as a quadratic equation in  $y$ . [2]

(ii) Hence or otherwise solve the equation  $3^{2x} + \frac{1}{3^{-x}} = 2$ . [6]



12 (a) Matrix  $N = \begin{pmatrix} 1 & 1 & 1 \\ x+1 & 1 & x \\ x & x+1 & 1 \end{pmatrix}$ , where  $x > 0$ .

Find the value of  $x$  given that  $|N| = 7$ . [3]

(b) Given the matrices  $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{pmatrix}$  and  $B = \begin{pmatrix} 0 & 2 & 2 \\ -2 & 0 & -1 \\ 1 & -1 & -1 \end{pmatrix}$

find

(i)  $AB$ , [3]

(ii)  $A^{-1}$ , [4]

(iii)  $B^{-1}$  [4]

(iv) Hence or otherwise, calculate  $(AB)^{-1}$ . [2]

